

## Research paper

# Assessment of CMIP6 GCMs for selecting a suitable climate model for precipitation projections in Southern Thailand

Usa Wannasingha Humphries<sup>a,1</sup>, Muhammad Waqas<sup>b,c,1</sup>, Phyo Thandar Hlaing<sup>b,c</sup>,  
Porntip Dechpichai<sup>a</sup>, Angkool Wangwongchai<sup>a,\*</sup>

<sup>a</sup> Department of Mathematics, Faculty of Science, King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

<sup>b</sup> The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

<sup>c</sup> Center of Excellence on Energy Technology and Environment (CEE), Ministry of Higher Education, Science, Research and Innovation, Bangkok, Thailand

## ARTICLE INFO

## Keywords:

Climate change  
CMIP6 GCMs  
Bias correction  
Precipitation projection  
Southern Thailand

## ABSTRACT

The selection of General Circulation Models (GCMs) is critical due to computational limitations and underlying uncertainties. This study provides a comprehensive assessment of three bias correction (BC) methods, namely the delta change method (DT), quantile mapping (QM), and empirical quantile mapping (EQM). Utilizing precipitation data from 30 observation stations across Southern Thailand, the evaluation encompasses five CMIP6 GCM models (CAM5-CSM1-0, CanESM5, CNRM-CM6-1, CNRM-ESM2-1, IPSL-CM6A-LR). Evaluation metrics, root mean square error (RMSE), mean absolute error (MAE), Pearson's correlation ( $r$ ), index of agreement ( $d$ ), and mean bias error (MBE) are employed to assess BC methods. Evaluation measures suggest that the DT method outperforms EQM and QM, with higher accuracy and lower errors (DT: RMSE = 3.61, MAE = 2.32; EQM: RMSE = 3.82, MAE = 3.70). Taylor diagrams show that CNRM-ESM2-1 has the highest correlation across sites ( $r = 0.36$ ), albeit with a wider dispersion, while CanESM5 has a more balanced performance. Significant annual precipitation increases are projected for different spans 2021–30, 2061–70, and 2091–2100, particularly under SSP585, which will influence flood risk, water management, and climate adaptation. Future projections with SSP585 continuously projecting a larger probability of increased precipitation. The DT method is recommended for the downscaling of CMIP6 GCMs for precipitation projections in Southern Thailand, recognizing its superior performance. The study's findings provide a foundation for informed decision-making and adaptation planning in southern Thailand, urging policymakers to prioritize climate resilience and adaptation strategies while using a multi-model ensemble approach for robust climate forecasts.

## 1. Introduction

Climate change (CC) is projected to impact the hydrological cycle and potentially affect water resource availability [1–4]. According to the Intergovernmental Panel on Climate Change (IPCC) assessments, droughts and global flooding are caused by CC [5]. Changes in hydro climatology may vary the frequency and intensity of severe precipitations likely to increase flooding [6]. According to the previous hydrological studies to measure and understand water cycle models must be broadened [7]. GCMs are extensively employed across various disciplines to make future climate projections, including hydrology and environmental studies [8]. In hydrology, GCMs frequently serve as input data for hydrological models, enabling assessments of CC effects on

runoff [9], sediment yield [10], precipitation, and temperature [11]. The 6th phase of the Coupled Model Intercomparison Project (CMIP6) introduces notable advancements compared to its predecessor CMIP5 [12]. These enhancements are attributed to quantifying radiative forcing, whether natural or anthropogenic, incorporating aerosol and land-use effects [13,14]. Other studies have also confirmed the added value of CMIP6 [15–18]. Many studies on the performance comparison between CMIP6 and CMIP5 within their specified study regions have produced different results. As, Li, Huo et al. (2021) stated that CMIP6 outputs demonstrate enhanced accuracy in precipitation statistics across different climatic regions, including Asian countries when compared to CMIP5 [11]. CC projections gain greater relevance and significance when they incorporate reference simulations [19]. It is also important to

\* Corresponding author.

E-mail address: [angkool.wan@kmutt.ac.th](mailto:angkool.wan@kmutt.ac.th) (A. Wangwongchai).

<sup>1</sup> These authors contributed equally to this work.

<https://doi.org/10.1016/j.rineng.2024.102417>

Received 1 February 2024; Received in revised form 3 June 2024; Accepted 11 June 2024

Available online 16 June 2024

2590-1230/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

acknowledge that model simulations may contain significant uncertainties, and focusing exclusively on CCs may lead to the oversight of systematic errors [11,20]. Due to an inadequate knowledge of the causing mechanisms of CC, CMIP6 GCM outputs often exhibit limitations in both spatial and temporal resolutions. These limitations can affect the accuracy of simulating climate characteristics at the catchment scale [21]. Therefore, it becomes imperative to downscale GCM outputs to make them suitable for hydrological applications. Moreover, utilizing methods like BC is crucial in refining climate projections to align more closely with observed data. These methods, which involve calibrating model outputs using historical observations, have been demonstrated to significantly enhance the accuracy of climate projections [21,22].

Two primary methodologies have emerged in the downscaling of GCM outputs: dynamic downscaling (DD) and statistical downscaling (SD). DD entails the integration of regional climate models (RCMs) within the framework of GCMs [23]. This method is characterized by its physical basis, independence from observational data, and applicability across various spatial resolutions. However, DD involves extensive computational demands and intricate model structures [24]. Conversely, according to Ref. [25], SD methods rely on the establishment of relationships between observed data (predictands) and large-scale variables (predictors) to predict CC [26]. This approach features relatively lower computational requirements and a simplified model formulation [27]. Furthermore, SD methods can rectify systematic errors inherent in GCMs [28]. SD operates on the initial assumption of future climate, implying that the empirical relationships established under current climate conditions remain applicable to future climates [11]. In addition to these advantages, SD is simple to implement, has a smaller negative impact on the surrounding environment, and calls for fewer resources [29]. In SD, selecting an optimal set of predictors for explaining the target variable must be fitted to specific predictors, geographical locations, seasons, and other relevant factors [30]. In general, the SD methods can be divided into three categories: regression (transfer function) method [31], stochastic weather generator [32], and weather pattern schemes. One of the most popular and common is BC, applied extensively for impact assessment and employed in CC studies worldwide [33,34].

Thailand, characterized by its tropical climate and agrarian economy, relies mainly on precipitation for its agricultural activities [2]. The southwest and northeast monsoons influence the climate in the country. Due to its geographical location, southern Thailand is susceptible to both natural climate variability and the potential impacts of CC [35]. Over 30 years, Thailand's climate exhibits an average annual temperature of 27 °C and an annual precipitation total of 1587.5 mm. However, the ongoing impact of CC has disrupted local climate patterns in the country. Between the late 20th and early 21st centuries, there was an evident increase of approximately 1 °C in the annual average temperature. Moreover, fluctuations in annual precipitation levels have resulted in both drought and flood scenarios [36]. According to the Long-Term Climate Risk Index for 2021, compiled by Germanwatch, Thailand ranked 9th out of 178 countries most affected by CC, witnessing a 146-fold increase in natural disasters over 21 years [37]. Therefore, this study is motivated by the imperative need for accurate climate projections to inform adaptation and mitigation strategies in the face of CC impacts. While previous studies have evaluated different SD and BC methods individually, our study offers a comprehensive comparative analysis of these methods within the context of GCM performance assessment. By interpreting the strengths and limitations of each BC method in refining GCM outputs, this study contributes to advancing the methodology for generating reliable climate projections in southern Thailand. Thus, this study aims to address the pressing need for improved accuracy and reliability in climate projections, thereby enhancing the understanding of future climate scenarios and informing effective decision-making processes.

The structure of the study is outlined as follows: The materials and

methods section which includes the geographical scope of the study, the compilation of station data, and the data sources employed, and offers a comprehensive overview of the research methodology. The subsequent section presents the outcomes of the BC methods and their respective performance evaluations. Furthermore, the assessment of GCM performance evaluates the performance of the reference dataset and explores the interrelationship between these elements across the study area. Extensive discussions are provided to elucidate the findings. The concluding section offers final insights and observations from the study, summarizing the key findings and their implications.

## 2. Methods and materials

### 2.1. Study area

Thailand is situated within the geographic coordinates of latitudes 5°37' to 20°27' N and longitudes 97°22' to 105°37' E, as illustrated in Fig. 1. Thailand experiences a tropical climate, resulting in consistently high temperatures, especially during March to May, with April registering as the hottest month with temperatures around 30 °C (86 °F) [38, 39]. Southern Thailand features a tropical climate and diverse topography, including coastal areas along the Andaman Sea to the west and the Gulf of Thailand to the east [40]. Both the southwest and northeast monsoons shape the climatic patterns. Given its geographical positioning, this region is vulnerable to natural fluctuations in climate and the potential consequences of CC [35].

### 2.2. Data collection and statistics

This study used daily observation precipitation data (predictands) from 30 Thai Meteorological Department (TMD) stations over the past three decades (1991–2022). This research ensured data quality by checking missing values before conducting a descriptive analysis. This study used the LSTM-RNN method, as Wangwongchai proposed (2023), to attribute any missing data [41]. Fig. 1 presents the rain gauge stations within the study area. The descriptive statistical analysis of precipitation data explains variances in mean precipitation across different geographical locations, as outlined in Table 1. For example, Petchaburi, Hua Hin, and Prachap Khirikhan have mean precipitation values of approximately 2.70 mm, 2.69 mm, and 3.28 mm, respectively, indicating low average precipitation levels. On the other hand, Ranong exhibits a significantly higher mean precipitation of 12.04 mm, suggesting a wetter climate in this area. Std is another crucial statistic, reflecting the degree of variability in precipitation data. Higher Std, such as those observed in Ranong (23.21 mm) and Nakhon Sithamarat (21.69 mm), indicate substantial variability in precipitation. In contrast, stations like Petchaburi and Hua Hin have lower Std (9.82 mm and 10.00 mm, respectively), suggesting more consistent precipitation patterns. Skewness (Cv) and kurtosis values provide insights into the shape and tail characteristics of the precipitation distribution. As seen in most stations, positive skewness indicates a right-skewed distribution, suggesting that the data is concentrated on the lower end of the precipitation range with occasional extreme values. Kurtosis values were extremely high in certain stations like Prachap Khirikhan (135.38), reflecting heavy tails in the distribution, signifying a higher likelihood of extreme precipitation events. The coefficient of variation (CV), expressed as a percentage, measures the relative variability of precipitation to its mean. Stations with high CV values, such as Prachap Khirikhan (329.70 %) and Ko Samui (298.51 %), experience significant variations in precipitation compared to their means. This highlights the need for understanding the factors contributing to this variability, which could be essential for local water resource management. The range between minimum and maximum precipitation values observed across stations reflects the diversity in precipitation patterns in the Southern Thailand region. Some stations, like Ranong and Nakhon Sithamarat, exhibit higher maximum precipitation values, implying a greater potential for heavy precipitation

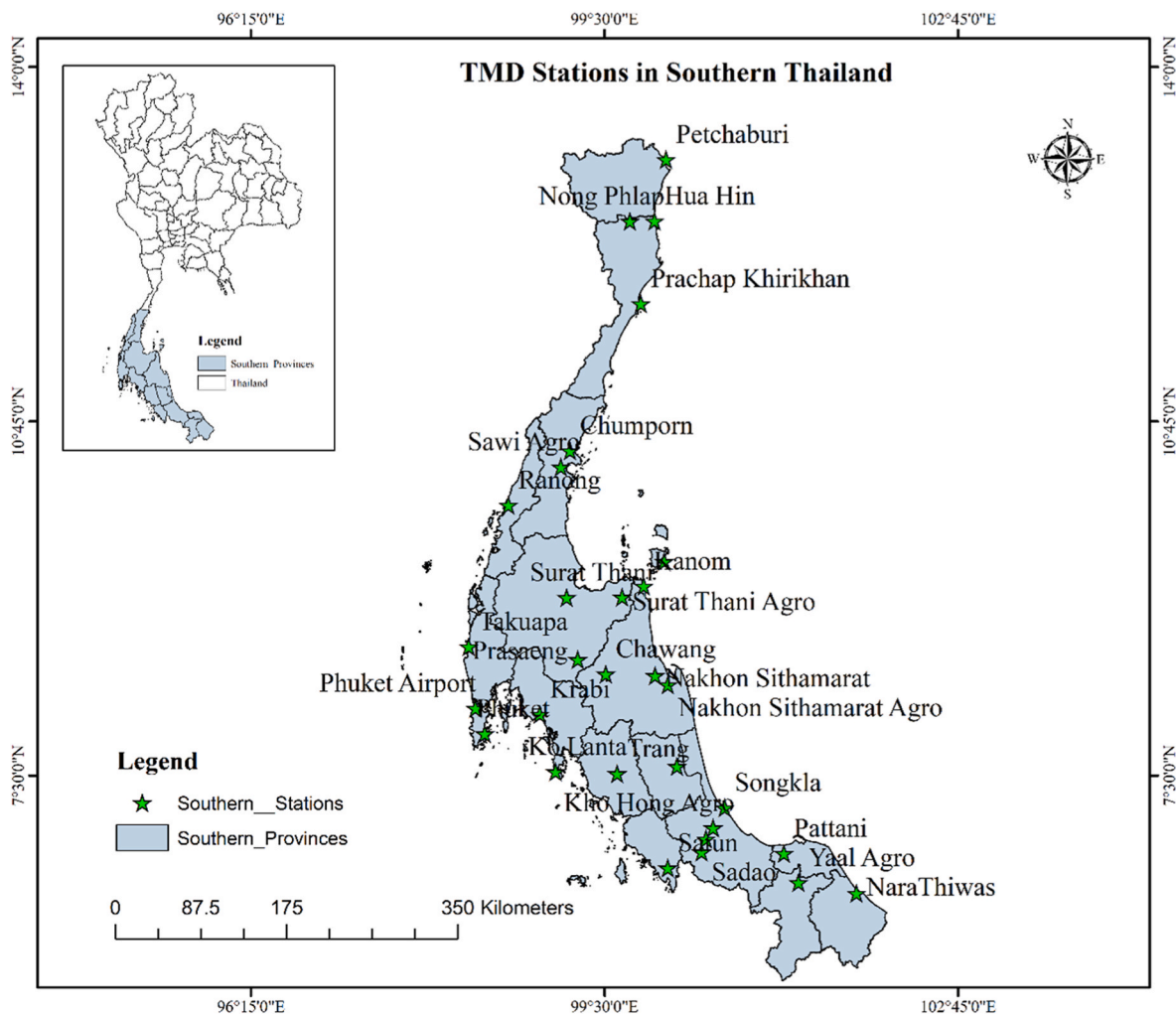


Fig. 1. Thai Meteorological Department rain gauge distribution over the southern region of Thailand.

events. In contrast, Petchaburi and Hua Hin have minimum precipitation values of zero, suggesting that these areas may experience periods with no recorded precipitation.

### 2.3. CMIP6-GCMs dataset

This study obtained data from the database of the World Climate Research Program (WCRP) CMIP6 GCM database, utilizing the daily ensemble of a single realization labeled as r1i1p1f1 variation. CMIP6 serves as a continuation and succession of CMIP5. The study selected five models from various institutions, specifically CAMS-CSM1-0, Can-ESM5, CNRM-CM6-1, CNRM-ESM2-1, and IPSL-CM6A-LR. Data from these CMIP6 GCMs (predictors) were obtained from the CMIP6 archives hosted by the Earth System Grid Federation (ESGF) WCRP, accessible through the link (<https://esgfnode.llnl.gov/projects/cmip6>) presented in Table 2. The selection of these five GCMs in this study is based on several key considerations and literature. Firstly, these GCMs represent a diverse set of institutions and research groups, ensuring a broad representation of modeling approaches and methodologies [42,43]. Secondly, these GCMs have been widely utilized in climate research and are recognized for their contributions to advancing the understanding of climate dynamics [44,45]. By including these specific models, this study aims to provide insights into the performance and reliability of some of the most commonly used GCMs in the context of climate projections for southern Thailand [46]. Additionally, the selection process considers the availability of data from these models in the CMIP6 archives, ensuring

accessibility and compatibility with research objectives. The study employs a comprehensive methodology for analyzing climate data in southern Thailand, as illustrated in Fig. 2. Data acquisition involves referencing the TMD dataset and accessing the World Climate Research Programme's CMIP6 GCMs database, covering scenarios SSP245, SSP370, and SSP585. Std follows, using BC techniques, DT, EQM, and QM. BC methods performance is rigorously assessed using metrics RMSE, MAE, MBE, correlation coefficient ( $r$ ), and ( $d$ ). The methodology includes a crucial step of selecting the most suitable CMIP6 GCM, determined through a detailed evaluation using a Taylor Diagram based on the Std Ratio and correlation coefficient ( $r$ ). Spatial projections under different SSPs for specific future periods are then generated.

### 2.4. Bias correction methods

After a comprehensive literature review presented in Table 3, the DT [52], QM [53], and EQM [54] methods were selected to downscale CMIP6 GCM models in the southern region of Thailand, which is characterized by its tropical climate and diverse geography, including coastal areas along the Andaman Sea to the west and the Gulf of Thailand to the east. The literature [47,52–56] fully explained these methods.

#### 2.4.1. Delta-change (DT)

The DT [57] creates climate scenarios not by directly using GCM simulations of the future but by adding the design research signal from

**Table 1**

Descriptive statistics of all rain gauges in the Southern region of Thailand for 1991–2022.

| Rain Gauges            | Longitude | Latitude | Mean  | Std   | Skewness | Kurtosis | C <sub>v</sub> | Min | Max    |
|------------------------|-----------|----------|-------|-------|----------|----------|----------------|-----|--------|
| Petchaburi             | 100.0667  | 13.15    | 2.70  | 9.82  | 9.31     | 155.08   | 364.08         | 0   | 282.20 |
| Prachap Khirikhan      | 99.83333  | 11.83333 | 3.28  | 10.81 | 9.00     | 135.38   | 329.70         | 0   | 296.90 |
| Hua Hin                | 99.96     | 12.58611 | 2.69  | 10.00 | 9.05     | 138.10   | 371.74         | 0   | 298.50 |
| Nong Phlap             | 99.73333  | 12.58333 | 3.24  | 9.49  | 7.59     | 96.59    | 293.01         | 0   | 226.00 |
| Chumporn               | 99.18333  | 10.48333 | 5.53  | 13.75 | 5.45     | 46.98    | 248.65         | 0   | 250.30 |
| Sawi Agro              | 99.1      | 10.33333 | 6.00  | 14.33 | 5.77     | 52.75    | 238.86         | 0   | 261.60 |
| Ranong                 | 98.61667  | 9.983333 | 12.04 | 23.21 | 3.19     | 12.97    | 192.69         | 0   | 208.00 |
| Surat Thani            | 99.15194  | 9.135556 | 4.52  | 11.86 | 8.91     | 136.94   | 262.07         | 0   | 296.90 |
| Ko Samui               | 100.05    | 9.466667 | 5.65  | 16.88 | 7.64     | 97.71    | 298.51         | 0   | 414.70 |
| Surat Thani Agro       | 99.66306  | 9.138889 | 5.82  | 15.89 | 7.78     | 93.36    | 273.03         | 0   | 323.80 |
| Prasaeng               | 99.255    | 8.565833 | 4.88  | 9.55  | 5.05     | 48.59    | 195.63         | 0   | 217.40 |
| Nakhon Sithamarat      | 99.96667  | 8.416667 | 7.55  | 21.69 | 7.05     | 75.16    | 287.19         | 0   | 405.00 |
| Kanom                  | 99.86111  | 9.239167 | 4.97  | 13.10 | 10.36    | 233.52   | 263.55         | 0   | 502.40 |
| Nakhon Sithamarat Agro | 100.0833  | 8.333333 | 7.27  | 19.70 | 7.89     | 125.35   | 270.93         | 0   | 615.60 |
| Chawang                | 99.51194  | 8.431944 | 5.34  | 10.22 | 4.63     | 35.78    | 191.28         | 0   | 195.00 |
| Pattalung Agro         | 100.1667  | 7.583333 | 6.30  | 17.31 | 6.22     | 59.71    | 274.55         | 0   | 312.80 |
| Takuapa                | 98.25222  | 8.684167 | 10.28 | 21.23 | 3.72     | 20.19    | 206.52         | 0   | 288.80 |
| Phuket                 | 98.4      | 7.883333 | 6.56  | 13.86 | 3.95     | 22.24    | 211.33         | 0   | 180.70 |
| Phuket Airport         | 98.31444  | 8.116667 | 7.63  | 16.54 | 4.31     | 27.85    | 216.64         | 0   | 245.70 |
| Ko Lanta               | 99.05     | 7.533333 | 6.66  | 15.25 | 4.58     | 31.57    | 228.99         | 0   | 241.60 |
| Krabi                  | 98.90667  | 8.0625   | 5.23  | 10.47 | 4.60     | 30.64    | 200.09         | 0   | 150.50 |
| Trang                  | 99.61667  | 7.516667 | 6.54  | 13.70 | 3.94     | 22.41    | 209.35         | 0   | 192.10 |
| Kho Hong Agro          | 100.5     | 7.016667 | 6.47  | 15.74 | 5.87     | 57.23    | 243.40         | 0   | 315.70 |
| Sadao                  | 100.3942  | 6.795556 | 4.16  | 10.13 | 5.98     | 56.72    | 243.69         | 0   | 187.70 |
| Songkla                | 100.6047  | 7.203889 | 6.36  | 18.80 | 8.33     | 124.02   | 295.77         | 0   | 521.80 |
| Hatyai                 | 100.4333  | 6.916667 | 5.05  | 12.41 | 5.41     | 47.58    | 245.84         | 0   | 219.40 |
| Satun                  | 100.0833  | 6.65     | 7.08  | 14.08 | 3.71     | 19.40    | 198.87         | 0   | 192.60 |
| Pattani                | 101.15    | 6.783333 | 5.77  | 15.36 | 6.47     | 65.64    | 266.37         | 0   | 310.50 |
| Yaal Agro              | 101.2833  | 6.516667 | 6.97  | 17.02 | 5.50     | 51.71    | 244.38         | 0   | 358.60 |
| NaraThiwas             | 101.8167  | 6.416667 | 8.21  | 21.76 | 7.09     | 90.11    | 265.08         | 0   | 545.40 |

**Table 2**

Attributes of CMIP6 general circulation models (GCMs).

| CMIP6-GCM    | Center              | Spatial resolution in degrees | Number of Ensemble Members | Developed country | SSP245 | SSP370 | SSP585 |
|--------------|---------------------|-------------------------------|----------------------------|-------------------|--------|--------|--------|
| CAMS-CSM1-0  | CAMS/China          | 1.1° × 1.1°                   | 3                          | China             | ◆      | ◆      | ◆      |
| CanESM5      | CCCMA/Canada        | 2.8° × 2.8°                   | 10                         | Canada            | ◆      | ◆      | ◆      |
| CNRM-CM6-1   | CNRM-CERFACS/France | 1.4° × 1.4°                   | 10                         | France            | ◆      | ◆      | ◆      |
| CNRM-ESM2-1  | CNRM-CERFACS/France | 1.4° × 1.4°                   | 1                          | France            | ◆      | ◆      | ◆      |
| IPSL-CM6A-LR | IPSL/France         | 2.5° × 1.3°                   | 20                         | France            | ◆      | ◆      | ◆      |

GCM predictions as a change to the observational datasets. This is done in place of using GCM projections of the future directly. Because of the following reasons:

$$P^{Mod}(t) = P^{obs}(t) \quad (1)$$

$$P^{frc}(t) = P^{obs}(t) \frac{[\mu d * P^{frc}(t)]}{[\mu d * P^{mod}(t)]} \quad (2)$$

"P" is the time series generated by the forecast that needs to be fixed, "t" is the time step, and "d" is the long-term daily mean. t is the only variable that has been changed. The time series known as P<sub>Mod</sub> was produced by the GCM simulations carried out during the control period, and the observations produced the time series known as P<sub>obs</sub>. On top of the observed time series are the GCM simulations of the variations between the controlled environment and the scenario. The process known as multiplicative correction is often carried out once each month. This is done to generate various potential outcomes for the projection of the future. During the calibration process, the DT will adjust the simulated time series of the GCM more precisely. On the other hand, due to the straightforward nature of the transfer, scheduled future shifts in dynamic climate range and unpredictability will not be considered.

#### 2.4.2. Empirical quantile mapping (EQM)

EQM uses a transfer function (TF) to adjust the distribution of the simulated data produced by the GCM to more closely match the distribution of the observational dataset [58]. The TF is utilized to achieve

this goal. The TF makes a comparison between the observed data and the data that was generated by the GCM based on statistical moments of the cumulative distribution function (CDF) for each function:

$$P^{Mod}(t) = CDF^{-1}_{obs}, m(CDF_{mod}, m(P^{obs}(t))) \quad (3)$$

$$P^{frc}(t) = CDF^{-1}_{obs}, m(CDF_{frc}, m(P^{frc}(t))) \quad (4)$$

While the extremes are maintained, the mean, variance, quantiles, and intensities are all subject to a non-linear correction that QM carries out. This correction is carried out while the extremes are preserved. As a direct result of this, the variability of the data that have been corrected now more closely resembles that of the data from the first GCM. With a precipitation threshold equivalent to PTR, the significant distributional distortion brought on by light precipitation can be reduced.

#### 2.4.3. Quantile mapping (QM)

It was introduced by Panofsky and Brier (1968); QM stands as a widely applied downscaling technique across various fields of study [59]. All methods of BC in GCM output rely on comparing the distribution of simulated climatic variables to that of locally observed data from the past. Furthermore, the primary implication in applying standard QM is the regularity assumption of the biases and other redistributive features, which means that these variables ought to be the same for the regard and future eras, while only the means of the allocation may change [60]. QM operates by comparing the cumulative distribution functions (CDFs) of GCM data with observed data. By aligning quantiles

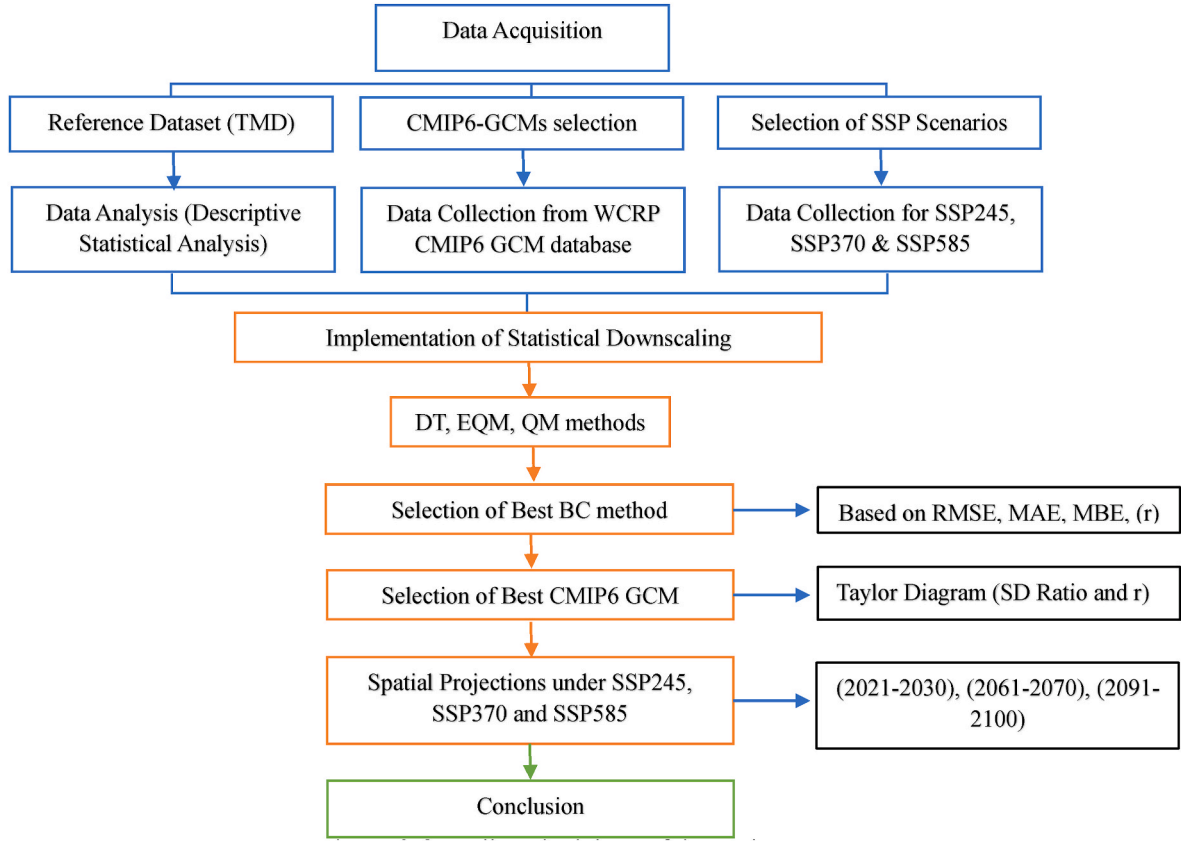


Fig. 2. Overall methodology of the study.

In both datasets, this method adjusts GCM projections, ensuring a closer match with the statistical characteristics of observed data. Consequently, this iterative process enhances the accuracy of downscaled climate information. The QM equation involves the calculation of the modeled probabilistic distribution to the observed probabilistic distribution. In the context of precipitation data, this concept is specifically computed in the equation below. For the evaluation of Standard Deviation (Std) GCM, Equation (5) is employed, while Equation (6) is utilized for future downscaling purposes.

$$P_{eva} = \text{InvCDF}_{Pr-Cal}^{stat} \left( \text{CDF}_{Pr-Cal}^{Hist} \left( p_{t-Eval}^{GCM} \right) \right) \quad (5)$$

$$P_{pre} = \text{InvCDF}_{Pr-Hist}^{stat} \left( \text{CDF}_{Pr-Hist}^{Hist} \left( p_{t-SSP}^{GCM} \right) \right) \quad (6)$$

The CDF represents the cumulative probability distribution of observational and General Circulation Model (GCM) data over a specified time.

### 2.5. Performance assessment

During the phase of performance evaluation, root means square error (RMSE), mean absolute error (MAE), Pearson's correlation (r), index of agreement (d), and the Mean bias error (MBE) are used to measure the bias between every BC method output and the observations.

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (P_{obs} - P_{pre})^2}{N}} \quad (7)$$

$$\text{Pearson's correlation } (r) = \frac{n(\sum P_{obs} * P_{pre}) - (\sum P_{obs})(\sum P_{pre})}{\sqrt{[n\sum P_{obs}^2 - (\sum P_{obs})^2][n\sum P_{pre}^2 - (\sum P_{pre})^2]}} \quad (8)$$

$$MBE = \frac{1}{n} \sum_{i=1}^n (P_{pre} - P_{obs}) \quad (9)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |P_{obs} - P_{pre}| \quad (10)$$

$$\text{Index of Agreement } (d) = 1 - \frac{\sum_{i=1}^N (P_{obs} - P_{pre})}{\sum_{i=1}^N (|P_{pre} - \overline{P_{obs}}| + |P_{obs} - \overline{P_{obs}}|)} \quad (11)$$

$P_{obs}$  denotes the observed precipitation,  $P_{pre}$  signifies the projected precipitation, and  $P_{mean}$  is the mean of the observed precipitation. Pearson's correlation coefficient (r) is a measure that ranges between  $-1.00$  and  $+1.00$ . A score of  $-1.00$  indicates a perfect negative correlation,  $+1.00$  signifies a perfect positive correlation, and a value of  $0.00$  indicates the absence of any relationship between the variables [61]. The index of agreement (d) ranges from 0 to 1, where higher values signify stronger agreement between the predicted values ( $P_{pre}$ ) and the observed values ( $P_{obs}$ ) [62]. The nominal value of RMSE indicates a good model [63]. MBE is employed to evaluate the model's average bias and to determine whether efforts are necessary to correct the bias. MBE quantifies the typical prediction bias. The MAE is the difference between the expected and observed values [59]. MAE and RMSE represent the average error in predicted values in units of the dependent variable, with values from 0 to  $\infty$  and independent of the error direction [64].

### 3. Results and discussions

This study employed five metrics to evaluate the accuracy of various bias correction (BC) methods. Additionally, Taylor diagrams were utilized to assess the performance of CMIP6 global circulation models

**Table 3**

Overview of recent investigations on bias corrections of CMIP6 and CMIP5 GCMs for the precipitation across varied geographic regions.

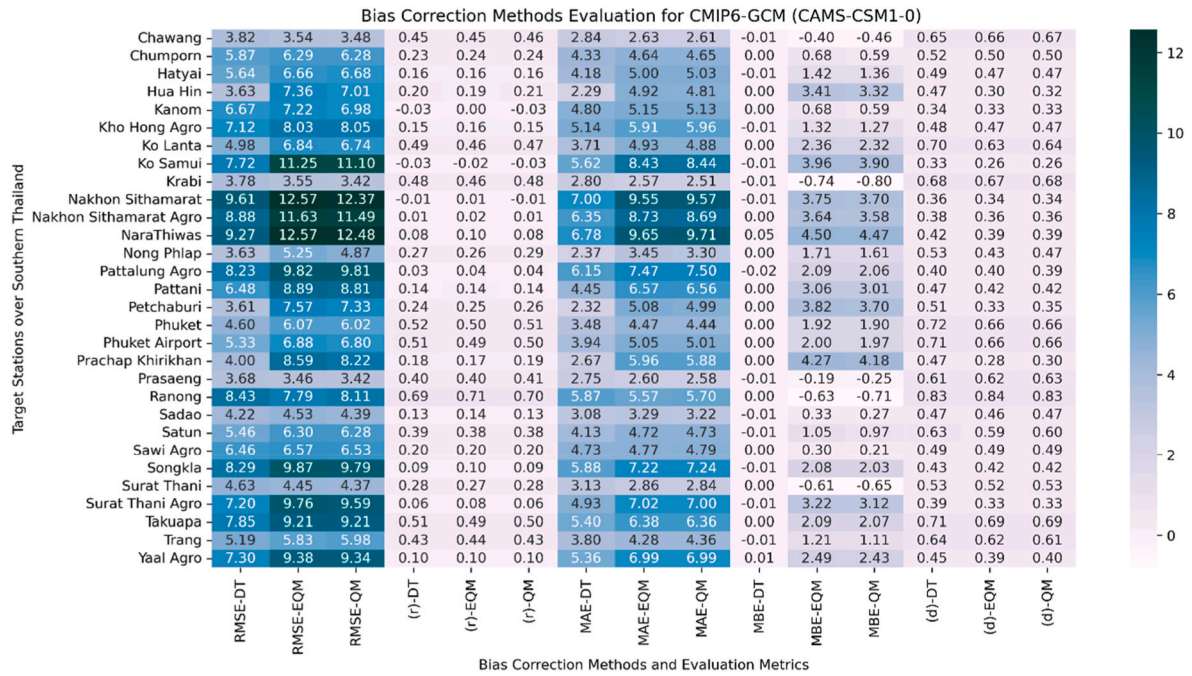
| Methods                                                                                                                                                                   | Historical period | GCM/RCM/Satellite Data                                                       | Region                          | Evaluation indices                                                                                                                                                                                                                                                | Key Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                            | Source |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Compromise Programming (CP) to calculate GCM's ranking                                                                                                                    | 1985–2014         | 24 CMIP6 GCMs                                                                | Tamil Nadu, India               | $R^2$ (Pearson correlation coefficient), NRMSE (Normalized Root Mean Square Error), PBIAS (Percentage Bias), and NSE (Nash–Sutcliffe Efficiency)                                                                                                                  | The study identifies suitable GCMs for predicting precipitation in different regions of Tamil Nadu during the North-East and Southwest monsoons, with CESM2 deemed appropriate for Chennai during both seasons.                                                                                                                                                                                                                                         | [22]   |
| DT, EQM, and QM                                                                                                                                                           | 1981–2010         | CMIP6 and CMIP5 GCMs                                                         | Upper Awash Basin, Ethiopia     | The evaluation is based on three criteria: (1) assessing the span of projected mean changes in climate variables, (2) examining the range of variability in climate extremes, and (3) appraising the model's proficiency in representing historical climate data. | This investigation made projections regarding temperature and precipitation changes for two scenarios, SSP4.5 and SSP8.5, across two specified study periods (2031–2060 and 2061–2090). The findings revealed spatial variations in the projections, indicating a decrease in precipitation downstream and an increase in mountainous regions, coinciding with elevated temperatures.                                                                   | [47]   |
| DT, EQM, and QM                                                                                                                                                           | 1965–2014         | CMIP6 GCM models (ACCESS-CM2, CNRM-CM6-1-HR, GFDL-ESM4, MIROC6, MRI-ESM2-0). | Türkiye                         | The assessment metrics include the correlation coefficient ( $r$ ) and the index of agreement ( $d$ ).                                                                                                                                                            | The study finds that DT performed better with observation data as a reference. The choice of reference data affected prediction accuracy, with a 65 % consistent performance achieved when using ERA reference data for BC.                                                                                                                                                                                                                             | [25]   |
| DT, EQM, and QM                                                                                                                                                           | 2003–2016.        | CMORPH, CFSR, PERSIAN, and TMPA_3B42V7 satellite products                    | Upper Blue Nile Basin, Ethiopia | $r$ , RMSE, NSE, MAE, MBE, and $d$                                                                                                                                                                                                                                | The study emphasized the importance of satellite precipitation products and BC methods in data-scarce regions for water resources planning and related sectors.                                                                                                                                                                                                                                                                                         | [48]   |
| Multi-criteria decision-making approach based on 4 BCE methods including Entropy, Gain Ratio, Symmetrical Uncertainty feature selection methods, Random Forest regression | 1971–2000         | 20 GCMs                                                                      | Nigeria                         | NRMSE, NSE, modified coefficient of agreement                                                                                                                                                                                                                     | MRI-CGCM3, HadGEM2-ES, CSIRO-Mk3-6-0, and CESM1-CAM5 were identified as the most suitable GCMs for Nigeria's rainfall projection. Linear scaling method preferred for downscaling, achieving NRMSE of 30.7–44.0 %, NSE 0.81–0.91, and MD 0.82–0.88. The projection indicates no significant rainfall change under RCP 2.6, 4.5, and 6.5, but a decrease under RCP 8.5 (2070–2099), with the seasonal increase in rainfall (0–20 %) in northern regions. | [49]   |
| LOCI, DT, and QM                                                                                                                                                          | 1961–2000         | 5-member ensemble CanESM2 and 10-member ensemble CSIRO-MK3.6                 | Hanjiang River (China)          | mean, standard deviation, and distribution                                                                                                                                                                                                                        | All BC methods applied resulted in a reduction of biases across all ensemble members.                                                                                                                                                                                                                                                                                                                                                                   | [50]   |
| Linear scaling, PT, and EQM                                                                                                                                               | 1983–2005         | GFDLES2G, MPI-ESMMR, MRI-CGCM3, GFDLCM3, and MRI-CGCM3                       | The central north of Egypt      | PDFs skill score                                                                                                                                                                                                                                                  | The results indicated that the LS (Least Squares) method outperformed other BC methods in addressing precipitation inaccuracies.                                                                                                                                                                                                                                                                                                                        | [51]   |

(GCMs) in simulating daily precipitation against observational data for the region. Following this evaluation, projections of annual precipitation were generated using the selected CMIP6 GCMs for the periods 2021–2030, 2061–2070, and 2091–2100.

### 3.1. Evaluation of bias correction methods through statistical metrics

The statistical evaluation of all BC methods for CMIP6-GCM (CAMS-CSM1-0) is shown in Fig. 3. At the Petchaburi station, the DT method yielded the lowest RMSE (3.61) and MAE (2.32), indicating a good fit to the observed data. It also showed a high correlation (0.24). The EQM

and QM methods had higher RMSE and MAE values and MBE values of 3.82 and 3.70, respectively, indicating substantial biases in these methods. The Index of Agreement ( $d$ ) for all BC methods at this station was similar, with DT showing a slightly better fit. Similar trends were observed at other stations, including Prachap Khirikhan, Hua Hin, Nong Phlap, Chumporn, Sawi Agro, and Ranong, where the DT method consistently outperformed EQM and QM in terms of RMSE, MAE, and MBE. It exhibited a better correlation with observed data ( $r$ ) and had a higher Index of Agreement ( $d$ ). Notably, at Ko Samui, DT also performed better regarding RMSE and MAE, though all methods showed low correlations. At some stations like Prasaeng, Pattalung Agro, Takuapa,

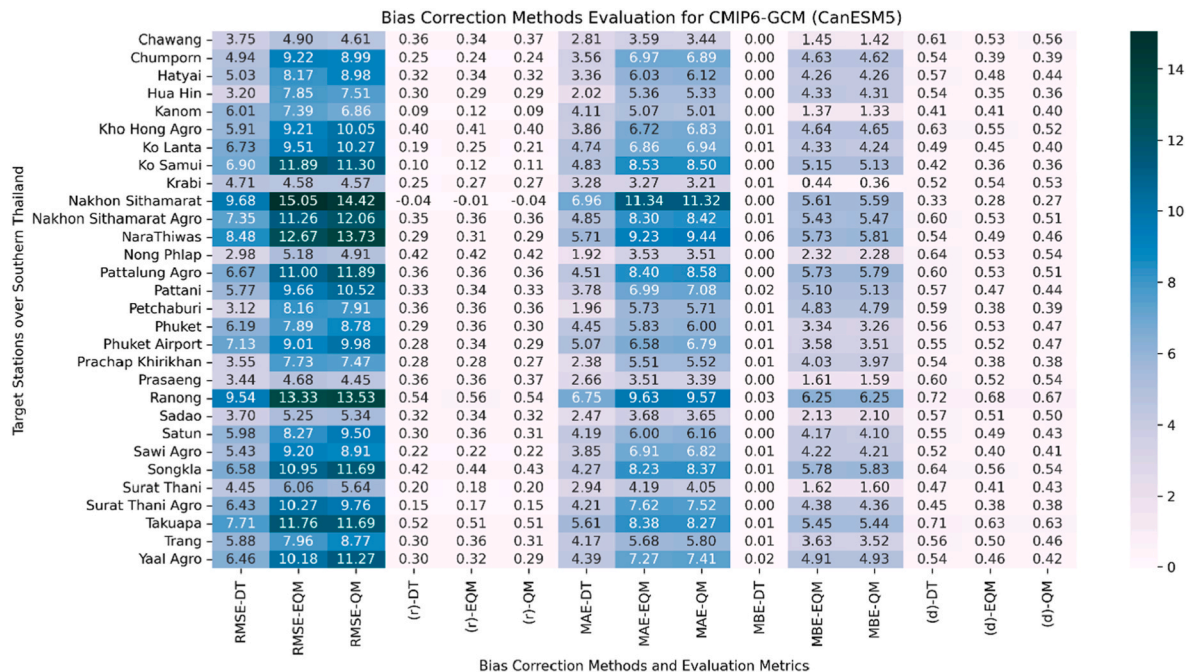


**Fig. 3.** Assessment of bias correction method performance (DT, EQM, QM) for CMIP6 GCM (CAMS-CSM1-0).

Phuket, Phuket Airport, Ko Lanta, and Krabi, EQM, and QM exhibited slightly better results in terms of RMSE and MAE. However, DT consistently displayed a higher correlation and a better Index of Agreement at most stations. Notably, at Ko Lanta, EQM and QM showed lower MBE values, suggesting a reduced bias in precipitation estimates compared to DT. At Nakhon Sithamarat, Kanom, Nakhon Sithamarat Agro, Chawang, and Sadao, the DT method performed better regarding RMSE and MAE and had a higher correlation ( $r$ ). It also showed a higher Index of Agreement ( $d$ ) at most stations, indicating better overall performance. However, at Songkla, Hatyai, and Satun, EQM, and QM displayed comparable results regarding RMSE, MAE, and MBE, with DT showing lower correlations and, in some cases, lower  $d$  values. In summary, the

DT method consistently outperformed EQM and QM in terms of RMSE, MAE, and MBE at most of the stations in the southern region of Thailand. It exhibited better correlations with observed data and a higher ( $d$ ).

The evaluation of BC methods for CanESM5 presented in Fig. 4 indicates some interesting patterns. At the Petchaburi station, the DT method exhibited the lowest RMSE (3.12) and MAE (1.96), suggesting a better fit to the observed data. It also had a high correlation ( $r = 0.36$ ), indicating a good match with the observations. While EQM and QM had higher RMSE and MAE values, they showed similar correlations, albeit with significant MBE values, indicating substantial biases in precipitation estimates. These trends continued at multiple stations, including Prachap Khirikhan, Hua Hin, Nong Phlap, Chumporn, Sawi Agro, and



**Fig. 4.** Assessment of bias correction method performance (DT, EQM, QM) for CMIP6 GCM (CanESM5).

Ranong, where the DT method consistently outperformed EQM and QM in terms of RMSE and MAE. It also demonstrated better correlations with observed data and had a higher Index of Agreement (d). At the Prasaeng station, all three methods performed well, with DT, EQM, and QM showing comparable RMSE, MAE, and MBE values. However, DT displayed a better correlation and a slightly higher Index of Agreement (d). Interestingly, at Nakhon Sithamarat, Kanom, Nakhon Sithamarat Agro, Chawang, and Pattalung Agro, EQM and QM exhibited better results in terms of RMSE and MAE, although DT continued to show a higher correlation and, in some cases, a better Index of Agreement. At Phuket, Phuket Airport, Ko Lanta, Krabi, Trang, Kho Hong Agro, Sadao, Songkhla, Hatyai, Satun, and Pattani, the DT method outperformed EQM and QM in terms of RMSE and MAE. In most cases, it also demonstrated better correlations and a higher Index of Agreement, indicating its suitability for these stations. For CanESM5 GCM data in the southern region of Thailand, the DT method consistently showed superior performance compared to EQM and QM in terms of RMSE, MAE, and MBE at most stations. It also had higher correlations and (d) values, suggesting its effectiveness in reducing biases in precipitation estimates.

For the CNRM-CM6-1 GCM, the results are shown in Fig. 5. Starting with the Petchaburi station, the DT method displays a low RMSE of 3.35 and MAE of 2.07, indicating a reasonable agreement with observed data. It also exhibits a moderate positive correlation ( $r = 0.36$ ), suggesting a connection with precipitation patterns. In contrast, the EQM and Quantile Mapping (QM) methods show higher RMSE and MAE values along with noticeable biases (MBE), indicating substantial discrepancies from the observed data at this station. Similar patterns are observed at the Prachap Khirikhan and Hua Hin stations, where the DT method consistently demonstrates a lower RMSE and MAE compared to EQM and QM.

The correlations with observed data are better with DT, implying a better match to the precipitation patterns. At the Nong Phlap station, DT again outperforms EQM and QM regarding RMSE, MAE, and MBE. This suggests that DT is more effective in reducing biases and aligning with observed data at this location. The Chumporn station displays a unique pattern where the DT method shows a notably higher RMSE than EQM and QM. However, it is important to note that DT maintains a moderate positive correlation and an acceptable Index of Agreement (d), indicating a reasonable fit to the observed data. In this case, EQM and QM

exhibit smaller RMSE and MAE values, but their correlations are comparatively weaker, suggesting some limitations. At Surat Thani and Ko Samui, the DT method presents a lower RMSE and MAE while maintaining a positive correlation. This suggests its effectiveness in mitigating biases and aligning with actual precipitation data. Across the remaining stations, including Phuket, Phuket Airport, Ko Lanta, Krabi, Trang, Kho Hong Agro, Sadao, Songkla, Hatyai, Satun, Pattani, Yaal Agro, and NaraThiwas, the DT method consistently displays competitive or superior performance in terms of RMSE, MAE, and MBE. It often exhibits positive correlations and higher (d) values, indicating reliability in reducing biases and aligning with the observed precipitation patterns. Overall, for CNRM-CM6-1 GCM data in the southern region of Thailand, the DT method tends to outperform EQM and QM in terms of aligning with observed precipitation patterns, even though it might occasionally result in slightly higher RMSE values.

For the GCM (CNRM-ESM2-1) at the Petchaburi station, the DT method exhibits low RMSE and MAE values, indicating a good agreement with the observed data shown in Fig. 6. It also shows a positive correlation ( $r = 0.35$ ) with actual precipitation patterns, making it a suitable choice for bias correction. In contrast, the EQM and QM methods show higher RMSE values and substantial biases (MBE), which may limit their accuracy. A similar pattern is observed at the Prachap Khirikhan and Hua Hin stations, where DT consistently outperforms EQM and QM regarding RMSE, MAE, and correlation with observed data. At the Nong Phlap station, DT again performs well, with lower RMSE and MAE compared to EQM and QM. It reduces biases and aligns better with actual precipitation data. The Chumporn station presents a unique pattern, with DT displaying a higher RMSE but maintaining a positive correlation with observed data. EQM and QM may offer smaller RMSE values but exhibit weaker correlations. Sawi Agro station shows that DT, despite higher RMSE, captures the variability in precipitation effectively, aligning well with observed data. The Ranong station presents a similar pattern, with DT having a stronger positive correlation, suggesting a trade-off between precision and alignment with observed data. At Surat Thani and Ko Samui, the DT method consistently offers lower RMSE and MAE values while maintaining positive correlations, indicating its effectiveness in reducing biases and aligning with actual precipitation patterns. Across the remaining stations, including Prasaeng, Nakhon Sithamarat, Kanom, Chawang, Pattalung Agro, Takuapa,

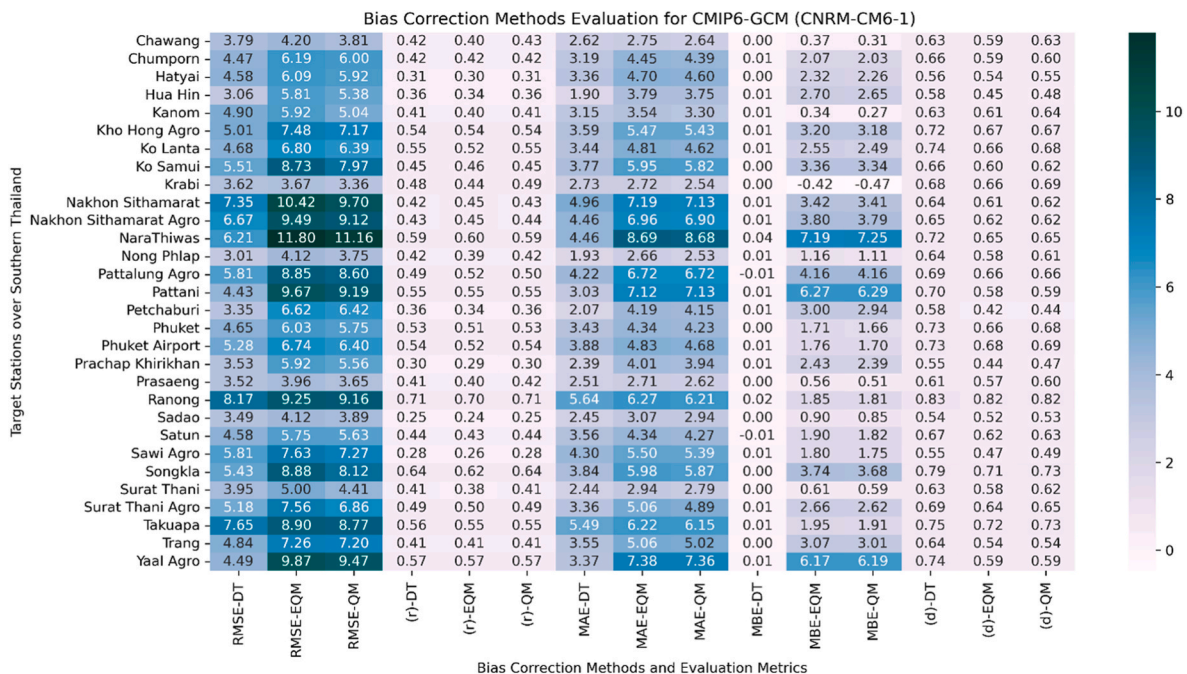
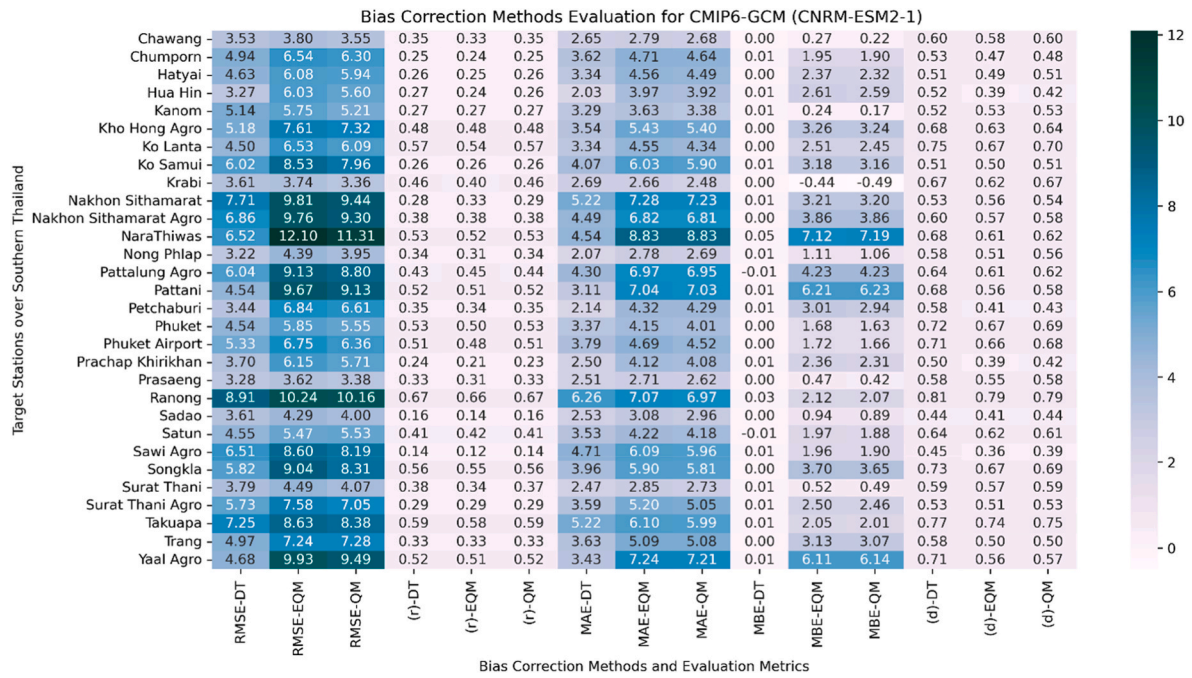


Fig. 5. Assessment of bias correction method performance (DT, EQM, QM) for CMIP6 GCM (CNRM-CM6-1).



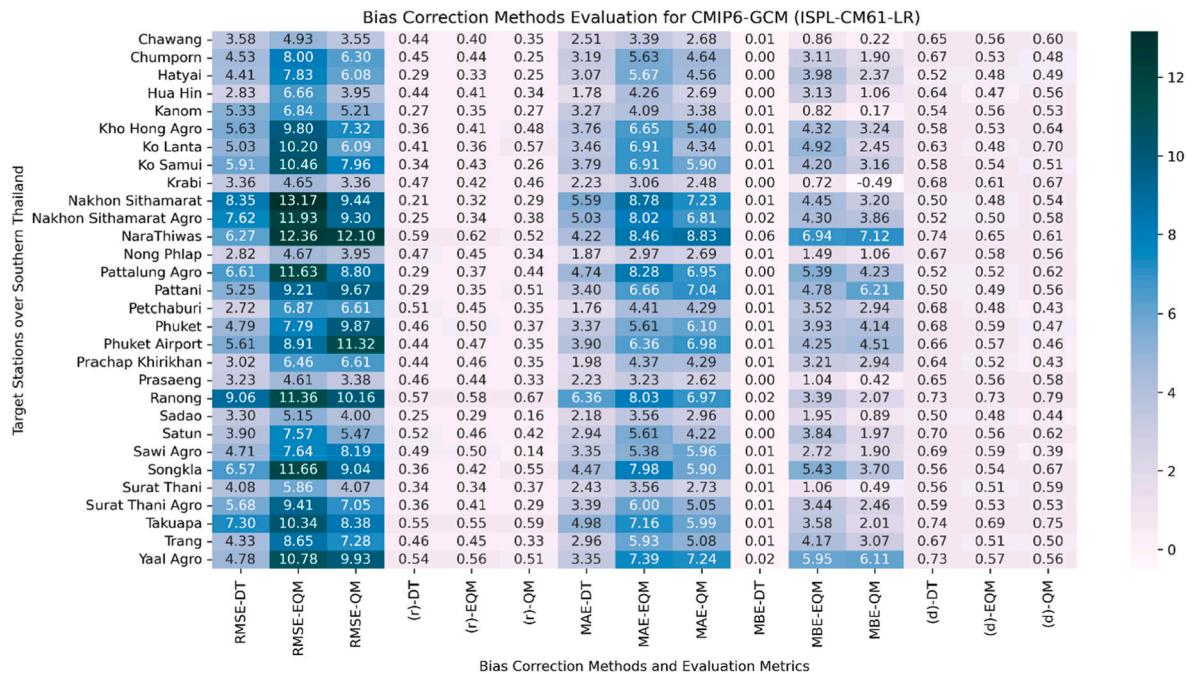
**Fig. 6.** Assessment of bias correction method performance (DT, EQM, QM) for CMIP6 GCM (CNRM-ESM2-1).

Phuket, Phuket Airport, Ko Lanta, Krabi, Trang, Kho Hong Agro, Sadao, Songkhla, Hatyai, Satun, Pattani, Yaal Agro, and NaraThiwas, DT generally exhibits competitive or superior performance in reducing biases and aligning with observed precipitation patterns, even though it may occasionally result in slightly higher RMSE values. In summary, for the GCM CNRM-ESM2-1 data in the southern region of Thailand, the DT method often outperforms the EQM and QM methods in terms of aligning with observed precipitation patterns.

Fig. 7 presents the results for IPSL-CM6A-LR. It is evident that the DT method consistently outperforms the EQM and QM across various regions. The RMSE, a crucial indicator of predictive accuracy, consistently favors DT with lower values in Petchaburi, Hua Hin, Nong Phlap, and

Chawang. Moreover, the Mean MAE supports the superiority of DT in most regions, except for specific cases where QM achieves slightly lower MAE values. The MBE also indicates that DT exhibits less bias overall, contributing to its robust performance. Additionally, (d) consistently favors DT with higher values across all regions, emphasizing its superior fit to the data. While the performance may vary across different stations, the overall trend suggests that DT is a more reliable method based on the provided statistical metrics, highlighting its potential as the preferred choice for these prediction tasks.

Across all six CMIP6 GCMs and various locations in the region, the DT consistently proves to be the superior BC method for precipitation projections. DT yields the lowest RMSE and MAE, demonstrating a



**Fig. 7.** Assessment of bias correction method performance (DT, EQM, QM) for CMIP6 GCM (IPSL-CM6A-LR).

robust fit to observed data. The method also exhibits a high correlation surpassing EQM and QM, showing higher RMSE and MAE values and significant MBE. This trend is consistently observed across multiple monitoring stations. Furthermore, the DT method consistently outperforms EQM and QM in terms of reducing biases, enhancing accuracy, and aligning with observed precipitation patterns across various CMIP6 GCMs and scenarios in the southern region of Thailand. Its reliable performance, as evidenced by all metric values, underscores its suitability for robust BC in the complex climate dynamics of the region.

### 3.2. Selection of the most effective CMIP6 GCM

To select suitable CMIP6 GCMs for the region Taylor diagrams are used to evaluate climate model daily precipitation simulations against observations [65]. The Taylor diagrams for the CMIP6 GCMs at all locations are presented in Fig. 8. CNRM-ESM2-1 consistently exhibits the highest correlation coefficient ( $r$ ) across various locations, achieving a perfect correlation ( $r = 1$ ) in several instances. This strong correlation underscores the reliability of the model's precipitation simulations to observed data. However, this analysis reveals that CNRM-ESM2-1 manifests a higher Std ratio, indicative of a broader dispersion of values compared to observed data. This observation suggests that although the model adeptly captures overarching precipitation patterns, there may be instances where simulated precipitation values deviate notably from observational records.

In contrast, CanESM5 consistently displays moderate ( $r$ ) and Std

ratios across the evaluated locations. The model balances accuracy and precision, reflecting a more stable representation of precipitation patterns. The ( $r$ ) values, while not as high as CNRM-ESM2-1, are still robust and, importantly, come with a more controlled spread of values. This suggests that CanESM5 provides a reliable and consistent precipitation simulation across diverse locations in southern Thailand. CAMS-CSM1-0, CNRM-CM6-1, and IPSL-CM6A-LR also exhibit moderate correlations and Std ratios. While not reaching the levels of correlation seen in CNRM-ESM2-1, these models maintain a reasonable balance between accuracy and precision. The consistent performance across multiple GCMs underscores the robustness of these models in representing the observed precipitation patterns in the study region. Considering the overall evaluation, the choice of the "best" GCM depends on the specific objectives and priorities of the study. Suppose the emphasis is on achieving a high correlation even at the expense of a wider spread of values. In that case, CNRM-ESM2-1 might be considered, and it emerges as a robust performer across multiple locations, demonstrating high consistency and reliability in simulating precipitation patterns.

### 3.3. Spatial projection of precipitation from selected CMIP6 GCM

In the comprehensive evaluation of CMIP6 GCMs, both CanESM5 and CNRM-ESM2-1 exhibited commendable performances. Building upon the reliable results of the previous assessment, spatial projections of annual precipitation were conducted utilizing these two selected GCMs under three scenarios (SSP345, SSP370, and SSP585) for three

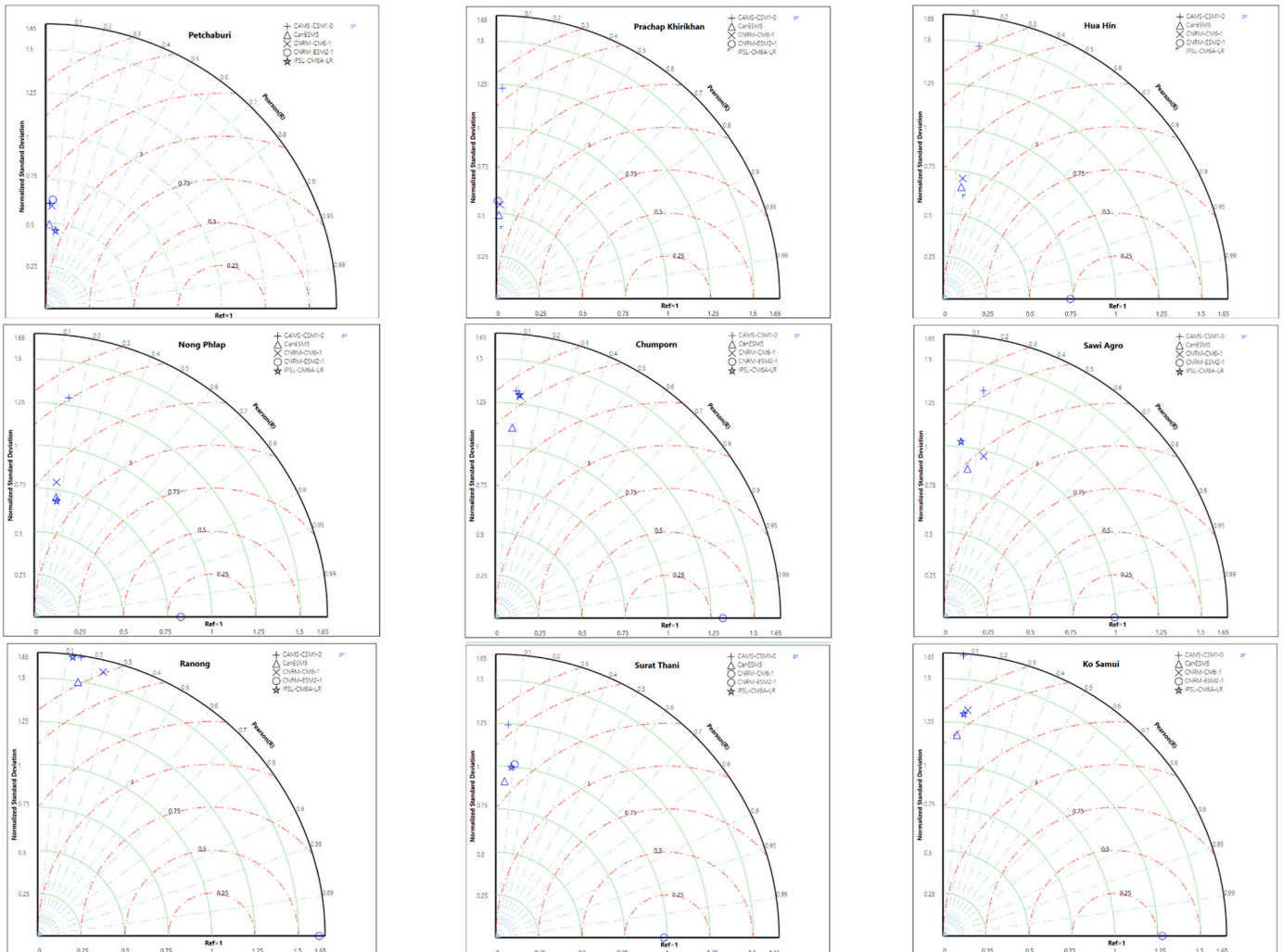


Fig. 8. Taylor Diagrams for the assessment of the CMIP6 GCMs at each location over southern Thailand.

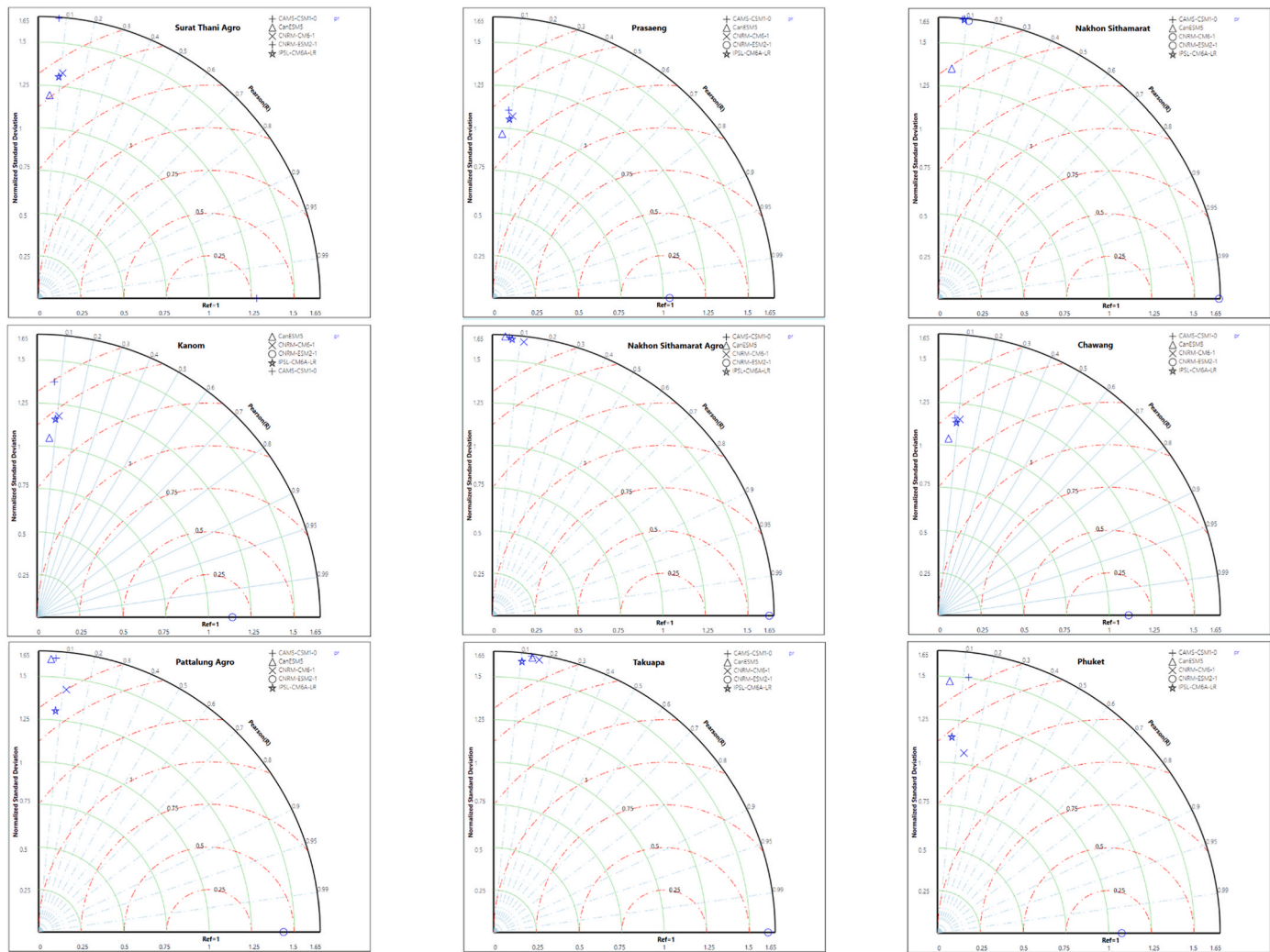


Fig. 8. (continued).

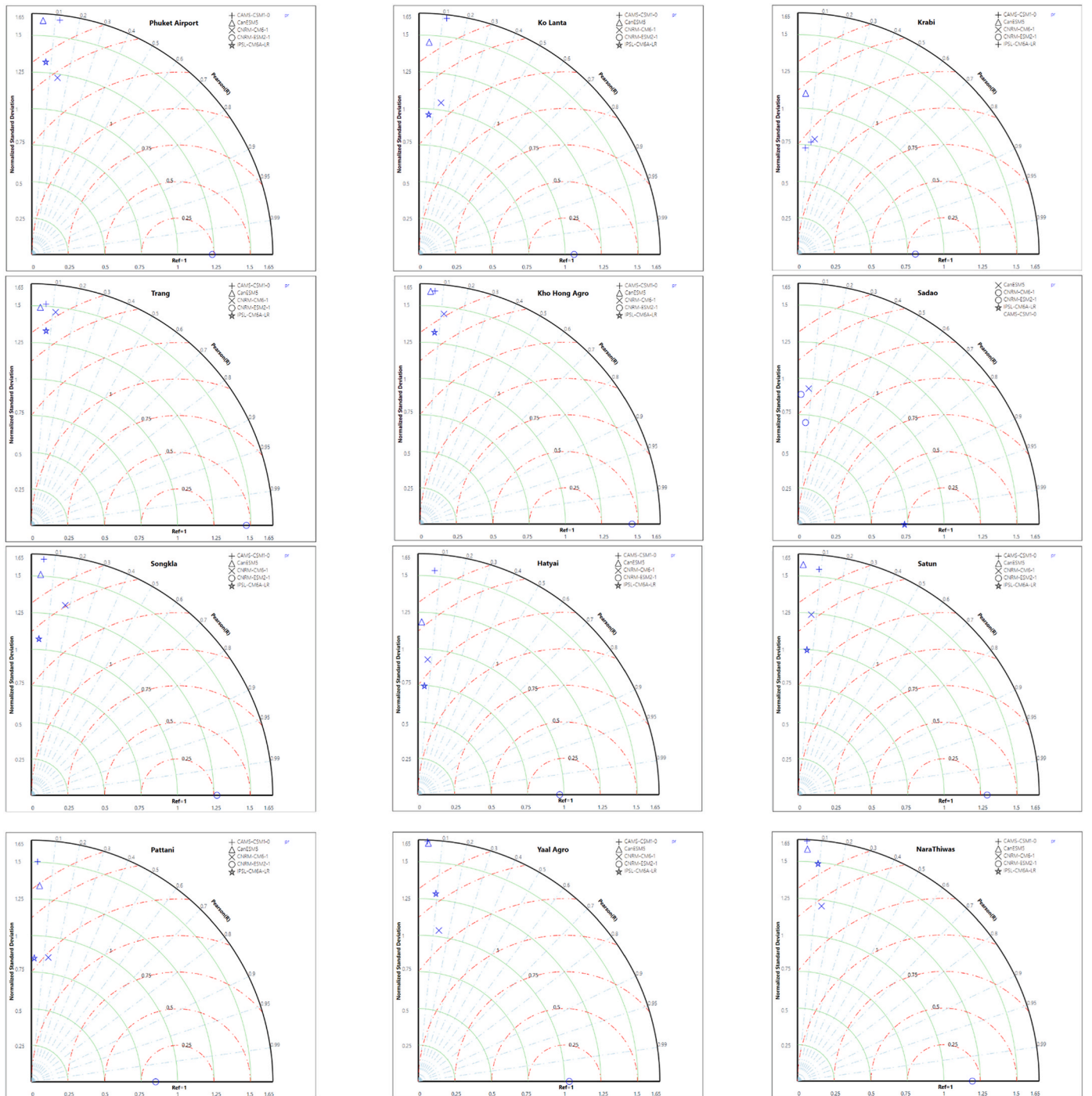


Fig. 8. (continued).

future periods: 2021–2030, 2061–2070, and 2091–2100. These projections are presented in Figs. 9 and 10.

Fig. 9 shows the annual precipitation projections simulated by the CanESM5 model under SSP245, SSP370, and SSP585. Under SSP245 for the period 2021–2030, precipitation amounts range from 10,000 to 17,000 mm annually, with certain areas nearing 20,000 mm. Moving to 2061–2070, projections indicate an increase in precipitation, with regions experiencing values between 17,000 and 27,000 mm annually, and some areas reaching up to 30,000 mm. The trend persists in 2091–2100, with widespread areas exhibiting precipitation surpassing 27,000 mm, and at some locations approaching 37,000 mm annually. For SSP370 and SSP585, the precipitation pattern in 2021–2030 mirrors

that of SSP245, with most regions experiencing precipitation ranging from 10,000 to 17,000 mm. However, in 2061–2070, a notable increase is observed, with many areas receiving between 17,000 mm and 27,000 mm, and some regions approaching the 30,000 mm threshold. By 2091–2100, a clear increase is seen with precipitation across much of the region ranging from 27,000 mm to 37,000 mm. The initial decade (2021–2030) does not reveal significant differences in precipitation amounts across all scenarios. SSP585 consistently projects the highest precipitation amounts, especially in the latter part of the century, suggesting that a high-emissions trajectory could lead to more precipitation. This is particularly evident in the 2091–2100 projections, where SSP585 suggests widespread regions receiving precipitation of more than

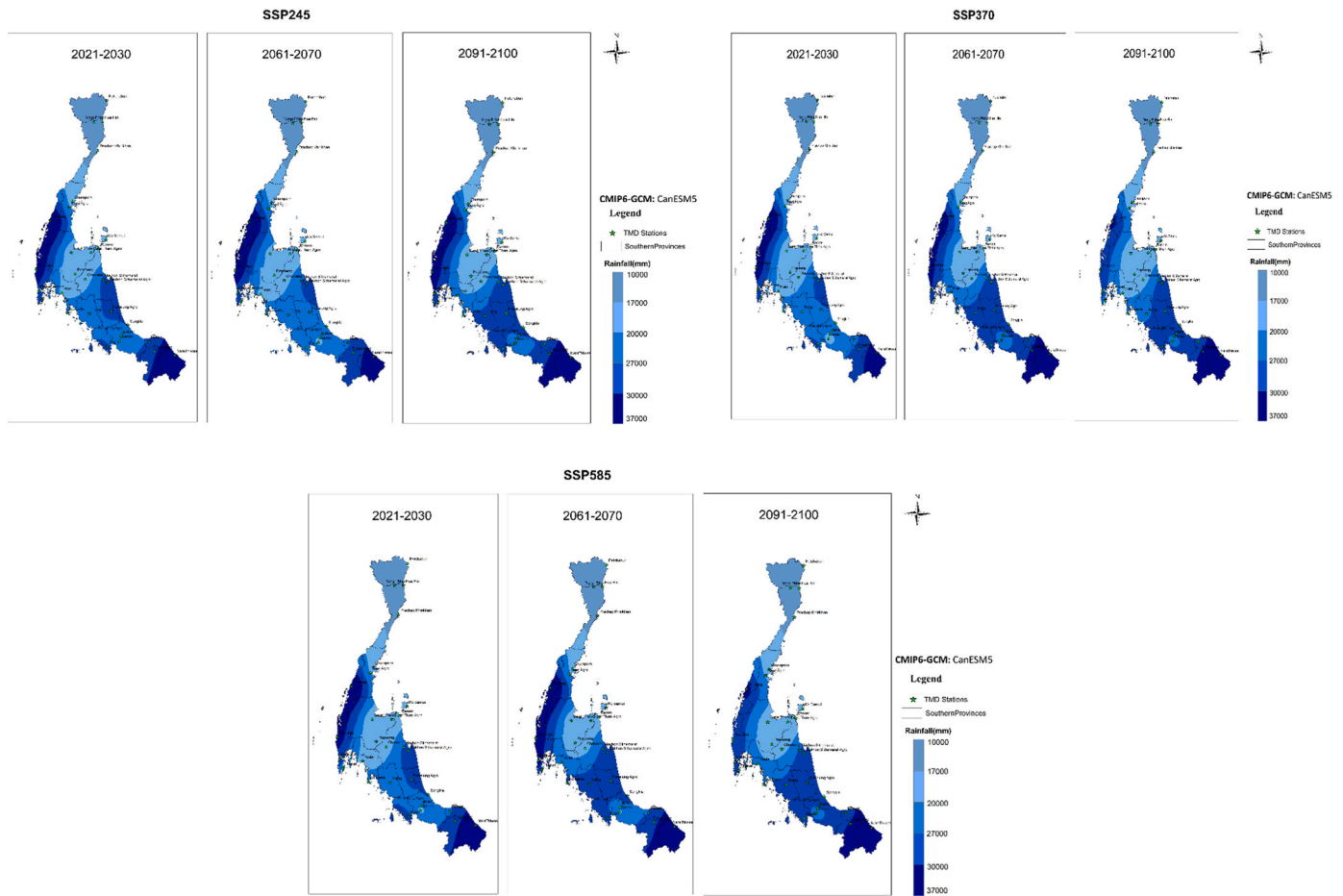


Fig. 9. Projection of precipitation with CanESM5 under SSP245, SSP370 and SSP585 for years (2021–2030), (2061–2070) and (2091–2100).

27,000 mm, reaching up to 37,000 mm, significantly higher than other scenarios.

Fig. 10 shows annual precipitation projections of CNRM-ESM2-1. In 2021–2030, SSP245 shows lighter shades, indicating lower precipitation. SSP370 exhibits slightly darker shades, suggesting increased precipitation. SSP585 mirrors SSP370, indicating similar precipitation patterns. In 2061–2070 and 2091–2100, SSP245 shows moderate increases with slightly darker shades. SSP370 indicates darker shades, signifying more precipitation, while SSP585 shows even darker shades, especially in central regions, implying the highest precipitation. In 2021–2030, SSP370 and SSP585 show more precipitation than SSP245. In the mid-term (2061–2070), SSP585 indicates a significant increase compared to SSP370 and SSP245. In the long term (2091–2100), SSP585 exhibits the most substantial increase, suggesting the highest precipitation under high-emission scenarios. These projections emphasize potential increases in precipitation and extreme weather events, impacting water management, agriculture, and infrastructure. Across SSP scenarios, precipitation is expected to increase over the century. SSP585 consistently shows the most significant increase, implying profound climate pattern changes, with severe implications for flood risk management, agriculture, and urban planning.

### 3.4. Discussion

The future climate outlook for southern Thailand predicts increased variability due to CC, posing significant challenges to water resources and exacerbating farmers' vulnerability to natural disasters. Consequently, it is imperative to conduct a detailed examination of anticipated changes in precipitation patterns within the CC framework for this

region. This study utilizes three BC methods to project future precipitation patterns in southern Thailand, employing CMIP6 GCMs despite acknowledged biases in model simulations. Results indicate a tendency for GCMs to overestimate lower precipitation amounts while underestimating extreme events, consistent with prior research on the observed drizzle effect in climate models [22,33,48,66].

The suitability of BC methods in tropical regions like southern Thailand is unexplored. Therefore, identifying appropriate BC methods becomes crucial for accurately assessing CC impacts on water resources. This research utilizes three BC methods (DT, EQM, and QM) to refine CMIP6 GCM outputs, enabling a comprehensive analysis of CC effects on precipitation patterns in southern Thailand. Despite the improvement introduced by BC methods, notable residual biases persist, displaying a notably non-linear nature. Remarkably, DT outperforms both EQM and QM across various evaluation indices, exhibiting statistical characteristics roughly comparable to observed data, aligning with previous studies [25,50], and emphasizing the importance of nuanced BC methods in tropical climate contexts. The DT method is applied to correct projected precipitation for three future periods (2021–2030, 2061–2070, and 2091–2100) under three scenarios (SSP245, SSP370, and SSP585). The study focuses on analyzing temporal and spatial changes in projected precipitation. Significant increases are anticipated across most TMD stations under most scenarios. For instance, in the 2091–2100 projections, the SSP585 scenario suggests widespread regions receiving precipitation exceeding 27,000 mm, reaching up to 37,000 mm, higher than other scenarios. This notable increase is particularly observed in the southern region, including Ranong, nearby provinces, and NaraThiwas and its surroundings.

Despite the importance of assessing precipitation projections in

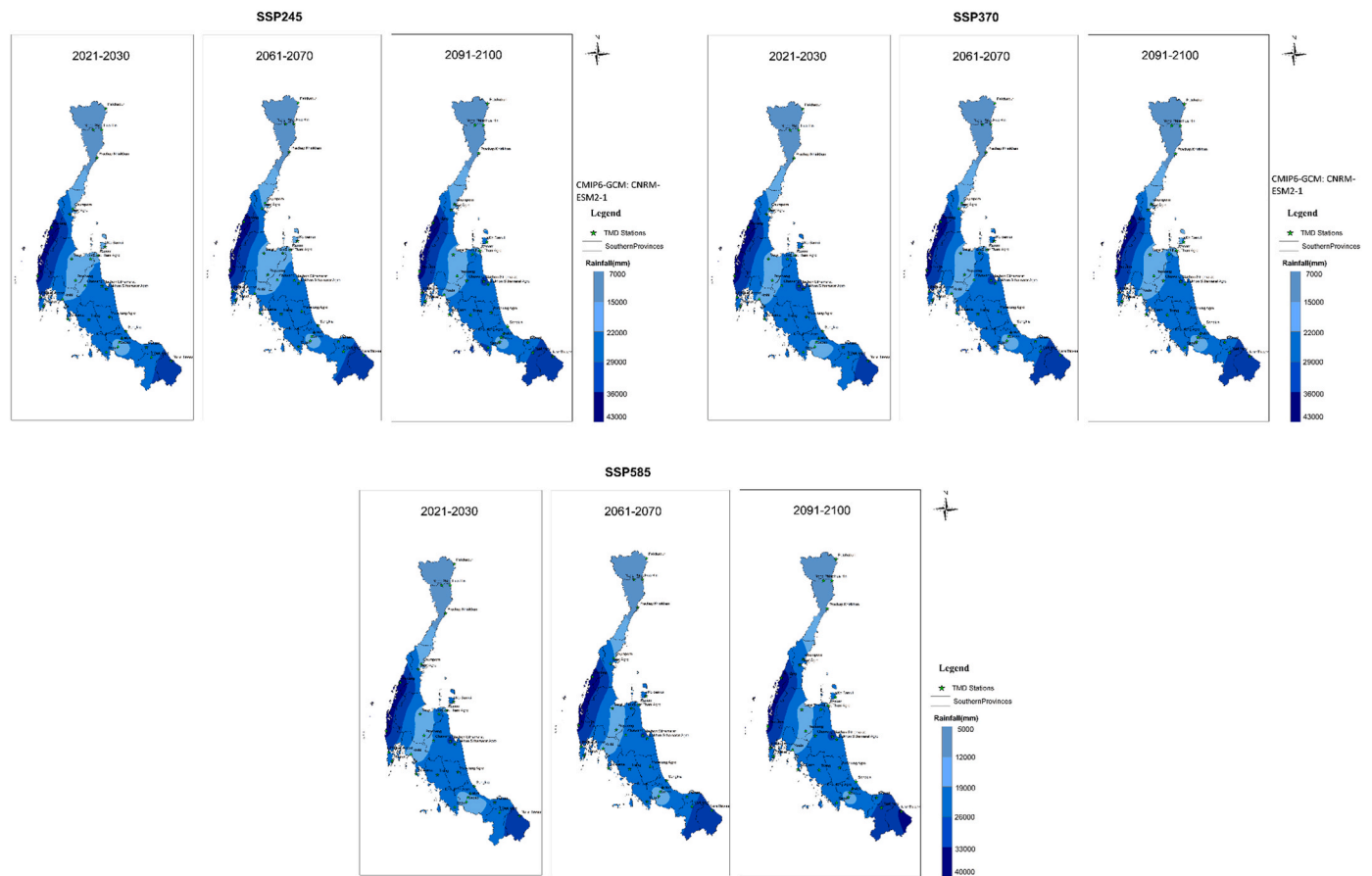


Fig. 10. Projection of precipitation with CNRM-ESM2-1 under SSP245, SSP370 and SSP585 for years (2021–2030), (2061–2070) and (2091–2100).

Thailand within the context of CC, a limited number of studies, such as those by Refs. [36,67–71], have been conducted. However, these studies have notable limitations concerning projection uncertainty. These include the utilization of coarse-gridded GCMs, a restricted number of gauged stations, and a dependency on gridded observation data for deriving correction factors for rainfall patterns projected under specific scenarios (RCP4.5 and RCP8.5). In contrast, the current study addresses these concerns by evaluating three BC methods to determine the most effective approach for correcting precipitation indices throughout southern Thailand. The selected method (DT) is subsequently applied to project precipitation under three SSPs, utilizing gauged data to estimate adjustment coefficients tailored to the most suitable CMIP6 GCM for the study area. Although direct comparisons of spatial patterns between this study and previous ones pose challenges, the current findings are consistent with the projections made by Ref. [36], indicating an increase in rainfall in the southern region of Thailand.

The projected changes in precipitation patterns in Southern Thailand under different scenarios can be attributed to several key atmospheric processes influenced by CC. These include alterations in regional circulation patterns, such as shifts in monsoonal dynamics and changes in the position of the Intertropical Convergence Zone (ITCZ), which significantly impact the spatial and temporal distribution of rainfall across the region [72,73]. Additionally, variations in sea surface temperatures (SSTs) in the surrounding oceans, including the Indian Ocean and the Pacific Ocean, and the associated teleconnection mechanisms, such as the El Niño-Southern Oscillation (ENSO), play a crucial role in modulating regional climate variability, including precipitation patterns in Southern Thailand [74–76]. Furthermore, land-atmosphere interactions, including changes in land use/land cover patterns, urbanization, and anthropogenic activities, contribute to alterations in regional climate patterns by influencing local-scale atmospheric

circulation, moisture fluxes, and surface energy balance [77,78]. By elucidating these underlying atmospheric mechanisms, this study provides valuable insights into the dynamics of regional CC and their implications for water resources, agriculture, and societal resilience in Southern Thailand."

#### 4. Conclusion

This study can be concluded as an evaluation of precipitation projections in southern Thailand, using CMIP6 GCMs under varying scenarios and employing different BC methods. Through detailed analysis, this research sheds light on the reliability of climate models and their implications for the region's climate resilience and adaptation strategies.

- The evaluation of BC methods (DT, EQM, QM) reveals DT consistently outperforms EQM and QM, demonstrating superior accuracy and lower errors (RMSE, MAE).
- Taylor diagrams show CNRM-ESM2-1 with the highest correlation across locations, albeit with a broader spread, and CanESM5 with balanced performance. When selecting the most effective GCM, researchers should consider the balance between correlation and spread.
- Spatial projections indicate significant increases in precipitation under SSP585, impacting flood risk, water management, and climate adaptation.

This study's findings provide a basis for informed decision-making and adaptation planning in southern Thailand, guiding sustainable and resilient development. The implications for policymakers are significant, particularly in flood risk management, agriculture, and

infrastructure planning, urging the prioritization of climate resilience and adaptation strategies. Policymakers must acknowledge the evolving nature of climate models and stay informed about advancements and uncertainties to make informed decisions. Employing a multi-model ensemble approach is recommended for robust climate projections, offering a comprehensive understanding of potential future scenarios.

## Funding

This research received no external funding.

## CRedit authorship contribution statement

**Usa Wannasingha Humphries:** Writing – original draft, Methodology, Investigation, Conceptualization. **Muhammad Waqas:** Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Phyo Thandar Hlaing:** Writing – original draft, Visualization, Software, Formal analysis. **Porntip Dechpichai:** Writing – review & editing, Supervision, Resources, Investigation, Formal analysis. **Angkool Wangwongchai:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgements

This research project is supported by King Mongkut's University of Technology Thonburi (KMUTT), Thailand Science Research and Innovation (TSRI), and the National Science, Research and Innovation Fund (NSRF) Fiscal year 2024.

## References

- [1] M. Waqas, et al., Advancements in daily precipitation forecasting: a deep dive into daily precipitation forecasting hybrid methods in the tropical climate of Thailand, *MethodsX* (2024) 102757.
- [2] M. Waqas, et al., Assessment of advanced artificial intelligence techniques for flood forecasting, in: 2023 International Multi-Disciplinary Conference in Emerging Research Trends (IMCERT), IEEE, 2023.
- [3] M. Waqas, et al., Assessment of advanced artificial intelligence techniques for streamflow forecasting in Jhelum river basin, *Pakistan J. Agric. Res.* 34 (2021) 580.
- [4] U.W. Humphries, et al., Determination of crop water requirements and potential evapotranspiration for sustainable coffee farming in response to future climate change scenarios, *Smart Agricultural Technology* 8 (2024) 100435.
- [5] V. Masson-Delmotte, et al., Climate change 2021: the physical science basis, in: Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change vol. 2, 2021.
- [6] F. Zhang, et al., What is the predictability limit of midlatitude weather? *J. Atmos. Sci.* 76 (4) (2019) 1077–1091.
- [7] B. Mohammadi, A review on the applications of machine learning for runoff modeling, *Sustainable Water Resources Management* 7 (6) (2021) 1–11.
- [8] A. John, et al., How to incorporate climate change into modelling environmental water outcomes: a review, *Journal of Water and Climate Change* 11 (2) (2020) 327–340.
- [9] A.D. Teklesadik, et al., Inter-model comparison of hydrological impacts of climate change on the Upper Blue Nile basin using ensemble of hydrological models and global climate models, *Climatic Change* 141 (2017) 517–532.
- [10] Y.D. Kim, J.M. Kim, B. Kang, Projection of runoff and sediment yield under coordinated climate change and urbanization scenarios in Doam dam watershed, Korea, *Journal of Water and Climate Change* 8 (2) (2017) 235–253.
- [11] H. Birara, R. Pandey, S.K. Mishra, Projections of future rainfall and temperature using statistical downscaling techniques in Tana Basin, Ethiopia, *Sustainable Water Resources Management* 6 (2020) 1–17.
- [12] B.C. O'Neill, et al., The scenario model intercomparison project (ScenarioMIP) for CMIP6, *Geosci. Model Dev. (GMD)* 9 (9) (2016) 3461–3482.
- [13] K. Wyser, et al., Warmer climate projections in EC-Earth3-Veg: the role of changes in the greenhouse gas concentrations from CMIP5 to CMIP6, *Environ. Res. Lett.* 15 (5) (2020) 054020.
- [14] S.Ç. Bağcı, et al., Intercomparison of the expected change in the temperature and the precipitation retrieved from CMIP6 and CMIP5 climate projections: a Mediterranean hot spot case, Turkey, *Atmos. Res.* 256 (2021) 105576.
- [15] M.S. Shiru, et al., Comparison of precipitation projections of CMIP5 and CMIP6 global climate models over Yulin, China, *Theor. Appl. Climatol.* 147 (2022) 535–548.
- [16] N. Khan, et al., Prediction of droughts over Pakistan using machine learning algorithms, *Adv. Water Resour.* 139 (2020) 103562.
- [17] M.S. Shiru, et al., Replicability of annual and seasonal precipitation by CMIP5 and CMIP6 GCMs over East Asia, *KSCE J. Civ. Eng.* 26 (4) (2022) 1978–1989.
- [18] M.S. Shiru, J.H. Kim, E.-S. Chung, Variations in projections of precipitations of CMIP6 global climate models under SSP 2–45 and SSP 5–85, *KSCE J. Civ. Eng.* 26 (12) (2022) 5404–5416.
- [19] U.W. Humphries, et al., Determination of crop water requirements and potential evapotranspiration for sustainable coffee farming in response to future climate change scenarios, *Smart Agricultural Technology* (2024) 100435.
- [20] N. Goldenson, et al., Using large ensembles to identify regions of systematic biases in moderate-to-heavy daily precipitation, *Geophys. Res. Lett.* 48 (9) (2021) e2020GL092026.
- [21] P. Hosseinzadehtalaei, et al., Climate change impact assessment on pluvial flooding using a distribution-based bias correction of regional climate model simulations, *J. Hydrol.* 598 (2021) 126239.
- [22] L. Vr, Performance evaluation of CMIP6 climate models for selecting a suitable GCM for future precipitation at different places of Tamil Nadu, *Environ. Monit. Assess.* 195 (8) (2023) 928, 928.
- [23] T. Engen-Skaugen, Refinement of dynamically downscaled precipitation and temperature scenarios, *Climatic Change* 84 (3–4) (2007) 365–382.
- [24] Z. Xu, Y. Han, Z. Yang, Dynamical downscaling of regional climate: a review of methods and limitations, *Sci. China Earth Sci.* 62 (2019) 365–375.
- [25] S. Oruc, Performance of bias corrected monthly CMIP6 climate projections with different reference period data in Turkey, *Acta Geophys.* 70 (2) (2022) 777–789.
- [26] E. Zorita, H. Von Storch, The analog method as a simple statistical downscaling technique: comparison with more complicated methods, *J. Clim.* 12 (8) (1999) 2474–2489.
- [27] D. Maraun, et al., Precipitation downscaling under climate change: recent developments to bridge the gap between dynamical models and the end user, *Rev. Geophys.* 48 (3) (2010).
- [28] I. Hanssen-Bauer, et al., Statistical downscaling of climate scenarios over Scandinavia, *Clim. Res.* 29 (3) (2005) 255–268.
- [29] R.L. Wilby, C.W. Dawson, E.M. Barrow, SDSM—a decision support tool for the assessment of regional climate change impacts, *Environ. Model. Software* 17 (2) (2002) 145–157.
- [30] A.H. Baghanam, et al., Assessing the impact of climate change over the northwest of Iran: an overview of statistical downscaling methods, *Theor. Appl. Climatol.* 141 (2020) 1135–1150.
- [31] S. Hadipour, et al., Transfer function models for statistical downscaling of monthly precipitation, *J. Teknol* 78 (2016) 4–9.
- [32] C.W. Richardson, Stochastic simulation of daily precipitation, temperature, and solar radiation, *Water Resour. Res.* 17 (1) (1981) 182–190.
- [33] M. Vrac, P.V. Ayar, Influence of bias correcting predictors on statistical downscaling models, *J. Appl. Meteorol. Climatol.* 56 (1) (2017) 5–26.
- [34] K. Salvi, S. Kannan, S. Ghosh, Statistical downscaling and bias-correction for projections of Indian rainfall and temperature in climate change studies, in: 4th International Conference on Environmental and Computer Science, 2011.
- [35] M. Waqas, et al., Potential of artificial intelligence-based techniques for rainfall forecasting in Thailand: a comprehensive review, *Water* 15 (16) (2023) 2979.
- [36] M.B. Masud, et al., Changes in climate extremes over North Thailand, 1960–2099, *J. Climatol.* 2016 (2016).
- [37] D. Eckstein, V. Künzel, L. Schäfer, The Global Climate Risk Index 2021, Germanwatch, Bonn, 2021.
- [38] R. Chokngamwong, L. Chiu, TRMM and Thailand daily gauge rainfall comparison, in: Preprints, 20th Conf. On Hydrology, Amer. Meteor. Soc. P. Atlanta, GA, 2006.
- [39] S. Sangkhaphan, Y. Shu, Impact of rainfall on agricultural growth in Thailand: evidence in farming and fishing activities, *J. Econ. Sustain. Dev.* 10 (16) (2019) 162–174.
- [40] B. Chansawang, et al., Efficient and consistent adaptive mesh generation for geophysical models: a case study over the Gulf of Thailand, *AIP Adv.* 14 (5) (2024).
- [41] A. Wangwongchai, et al., Imputation of missing daily rainfall data; A comparison between artificial intelligence and statistical techniques, *MethodsX* 11 (2023) 102459.
- [42] M. Kamruzzaman, et al., Assessment of CMIP6 global climate models in reconstructing rainfall climatology of Bangladesh, *Int. J. Climatol.* 42 (7) (2022) 3928–3953.
- [43] M. Kamruzzaman, et al., Comparison of CMIP6 and CMIP5 model performance in simulating historical precipitation and temperature in Bangladesh: a preliminary study, *Theor. Appl. Climatol.* 145 (2021) 1385–1406.
- [44] N. Luo, et al., Added value of CMIP6 models over CMIP5 models in simulating the climatological precipitation extremes in China, *Int. J. Climatol.* 42 (2) (2022) 1148–1164.
- [45] F. Yazdandoost, et al., Evaluation of CMIP6 precipitation simulations across different climatic zones: uncertainty and model intercomparison, *Atmos. Res.* 250 (2021) 105369.

- [46] K. Tedprasith, W. Lohpaisankrit, Development of intensity–duration–frequency relationships in Khon Kaen City, Thailand under changing climate using GCMs and a simple scaling method, *Journal of Water and Climate Change* 15 (3) (2024) 1204–1217.
- [47] S.H. Gebresellase, et al., Evaluation of CMIP6 Climate Models for Climate Change Impact Assessments in Upper Awash Basin, Ethiopia, 2022.
- [48] A.B. Nigussie, et al., Evaluation of multiple bias correction methods with different satellite rainfall products in the main beles watershed, upper blue Nile (abbay) basin, Ethiopia, *Journal of Water and Climate Change* 14 (1) (2023) 156–174.
- [49] M.S. Shiru, et al., A MCDM-based framework for selection of general circulation models and projection of spatio-temporal rainfall changes: a case study of Nigeria, *Atmos. Res.* 225 (2019) 1–16.
- [50] J. Chen, et al., Bias correcting climate model multi-member ensembles to assess climate change impacts on hydrology, *Climatic Change* 153 (2019) 361–377.
- [51] M.S. Nashwan, S. Shahid, E.-S. Chung, High-resolution climate projections for a densely populated Mediterranean region, *Sustainability* 12 (9) (2020) 3684.
- [52] L.E. Hay, R.L. Wilby, G.H. Leavesley, A comparison of delta change and downscaled GCM scenarios for three mountainous basins in the United States 1, *JAWRA Journal of the American Water Resources Association* 36 (2) (2000) 387–397.
- [53] A.W. Wood, et al., Hydrologic implications of dynamical and statistical approaches to downscaling climate model outputs, *Climatic Change* 62 (1–3) (2004) 189–216.
- [54] J. Boé, et al., Statistical and dynamical downscaling of the Seine basin climate for hydro-meteorological studies, *Int. J. Climatol.: A Journal of the Royal Meteorological Society* 27 (12) (2007) 1643–1655.
- [55] Y.N. M'Po, et al., Comparison of daily precipitation bias correction methods based on four regional climate model outputs in Ouémé Basin, Benin, *Hydrology* 4 (6) (2016) 58–71.
- [56] X. Yang, et al., Bias correction of historical and future simulations of precipitation and temperature for China from CMIP5 models, *J. Hydrometeorol.* 19 (3) (2018) 609–623.
- [57] O. Rätty, J. Räisänen, J.S. Ylhäisi, Evaluation of delta change and bias correction methods for future daily precipitation: intermodel cross-validation using ENSEMBLES simulations, *Clim. Dynam.* 42 (2014) 2287–2303.
- [58] K. Byun, A.F. Hamlet, An improved empirical quantile mapping procedure for bias correction of climate change projections, *AGU Fall Meeting Abstracts* (2019).
- [59] C. Jacovides, H. Kontoyiannis, Statistical procedures for the evaluation of evapotranspiration computing models, *Agric. Water Manag.* 27 (3–4) (1995) 365–371.
- [60] M. Jakob Themeßl, A. Gobiet, A. Leuprecht, Empirical-statistical downscaling and error correction of daily precipitation from regional climate models, *Int. J. Climatol.* 31 (10) (2011) 1530–1544.
- [61] A. Wangwongchai, et al., Imputation of missing daily rainfall data; A comparison between artificial intelligence and statistical techniques, *MethodsX* (2023) 102459.
- [62] C.J. Willmott, On the evaluation of model performance in physical geography, *Spatial statistics and models* (1984) 443–460.
- [63] M. Shoaib, et al., A comparative study of various hybrid wavelet feedforward neural network models for runoff forecasting 32 (1) (2018) 83–103.
- [64] C.J. Willmott, K. Matsuura, Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance, *Clim. Res.* 30 (1) (2005) 79–82.
- [65] K.E. Taylor, *Taylor Diagram Primer*, 2005, pp. 1–4. Work. Pap.
- [66] V. Mishra, U. Bhatia, A.D. Tiwari, Bias-corrected climate projections for South Asia from coupled model intercomparison project-6, *Sci. Data* 7 (1) (2020) 338.
- [67] H. Kang, et al., Statistical downscaling of pattern projection using multi-model output variables as predictors, *Acta Meteorol. Sin.* 25 (3) (2011) 293–302.
- [68] P. Aksornsingchai, C. Srinilta, Statistical downscaling for rainfall and temperature prediction in Thailand, in: *Proceedings of the International Multiconference of Engineers and Computer Scientists*, 2011.
- [69] M.T. Vu, et al., Statistical downscaling rainfall using artificial neural network: significantly wetter Bangkok? *Theor. Appl. Climatol.* 126 (2016) 453–467.
- [70] H. Kang, et al., Multimodel output statistical downscaling prediction of precipitation in the Philippines and Thailand, *Geophys. Res. Lett.* 34 (15) (2007).
- [71] T.A. Gado, R.M. El-Hagrasy, I.M. Rashwan, Projection of rainfall variability in Egypt by regional climate model simulations, *Journal of Water and Climate Change* 13 (8) (2022) 2872–2894.
- [72] L. Huggi, et al., Climate change-induced spatiotemporal monsoon variability: assessment of its implications on global and regional production systems, in: *Climate Change Impacts on Soil-Plant-Atmosphere Continuum*, Springer, 2024, pp. 713–758.
- [73] S. Dhame, et al., Indian Ocean warming modulates global atmospheric circulation trends, *Clim. Dynam.* 55 (7) (2020) 2053–2073.
- [74] P. Kumar, et al., Influence of climate variability on sea level rise and its teleconnection with sea surface temperature anomalies over the Indo-Pacific Ocean, *Int. J. Climatol.* 42 (16) (2022) 10195–10216.
- [75] K. Power, et al., The role of El Niño in driving drought conditions over the last 2000 years in Thailand, *Quaternary* 3 (2) (2020) 18.
- [76] N.W. Ismail, S.M. Chan, Impacts of the El Niño-southern oscillation (ENSO) on paddy production in southeast Asia, *Clim. Dev.* 12 (7) (2020) 636–648.
- [77] M. Barlage, F. Chen, Human impacts on land surface-atmosphere interactions, *Fast Processes in Large-Scale Atmospheric Models: Progress, Challenges, and Opportunities* (2023) 213–228.
- [78] Q. Cao, et al., Impacts of landscape changes on local and regional climate: a systematic review, *Landsc. Ecol.* 35 (2020) 1269–1290.